

boomsonic_v0 — 500 N turbojet

Engine specification · single-spool, centrifugal compressor, axial turbine · case ce75000_opr4_t1150_b15_cap · generated 2026-09-14 from the project data

Status. Conceptual design study (student learning project). The numbers come from verified tool runs and cited relations: Phase 3R/4R analysis, the Phase 5 freeze and the Phase 6 CAD / 3D checks. **Nothing has been measured on hardware.** Performance is at the fielded technology level (component efficiencies that reproduce commercial micro-turbojet fuel burn) unless marked tool. **Open risks (section 9) are not resolved**, the top one being the compressor surge margin, which no validated method has predicted.

1. Principal data

parameter	value	condition / note
net thrust	500 N	design point M 1.02, 5 km ISA (sizing point)
max static thrust	631 N	sea level, M 0, 98.2 % speed, limited by T4 (100 % speed would need 1182 K)
air mass flow	1.326 kg/s	design point (corrected 1.362 kg/s); 1.330 kg/s at sea-level static
compressor pressure ratio	4.0	total-to-total, single centrifugal stage
turbine inlet temperature T4	1150 K (877 °C)	design limit; max thrust is T4-limited
specific fuel consumption	0.164 kg/(N·h)	design point; 0.136 kg/(N·h) at sea-level static
fuel flow	22.8 g/s (82 kg/h)	design point; 23.8 g/s at sea-level static
spool speed	75,000 rpm	max continuous (MCS) 78,750 rpm (105 %)
envelope (Ø × length)	186.6 × 426 mm	drawn components 352.5 mm long; the rest is calibrated length
dry mass	6.85 kg	calibrated (range 6.49–7.21); + 1.60 kg accessories (ECU, pump ...)
thrust / dry weight	9.4	sea-level static

2. Cycle at the design point

Flight M 1.02 at 5000 m ISA: ambient 255.6 K, 54.02 kPa, flight speed 327 m/s. The engine is sized for 500 N here; the gross thrust must also cover the ram drag of taking the air on board at flight speed.

quantity	fielded (design basis)	tool level	unit
net thrust	500	500	N
gross thrust / ram drag	934 / 434	849 / 349	N
air mass flow (corrected)	1.326 (1.362)	1.068 (1.101)	kg/s
specific thrust	377	468	N/(kg/s)
compressor pressure ratio	4.00	4.00	–
turbine inlet temperature T4	1150	1150	K
fuel flow / TSFC	22.8 / 0.164	19.2 / 0.138	g/s / kg/(N·h)
fuel-air ratio	0.0163	0.0171	–
compressor η_{is} / turbine η_{tt} / combustion η_b	0.70 / 0.75 / 0.95	0.817 / 0.936 / 0.95	–
compressor power	285	197	kW
turbine pressure ratio (tt)	2.60	1.88	–
nozzle: throat area / pressure ratio / jet velocity	70.8 / 2.76 / 566	41.9 / 3.81 / 573	cm ² / – / m/s
intake capture area	55.1	44.4	cm ²

Gas-path stations (fielded, total conditions)

station	total temperature	total pressure	note
0 ambient (static)	255.6 K	54.02 kPa	ISA at 5000 m; flight speed 327 m/s
2 compressor inlet	308.9 K	102.1 kPa	after the pitot intake (normal shock + subsonic duct loss)
3 compressor exit	520.3 K	408.4 kPa	
4 turbine inlet	1150 K	388.0 kPa	combustor pressure loss 5.0 %
5 turbine exit	971.5 K	149.3 kPa	698 °C

3. Sea-level static (take-off) point

thrust	air flow	fuel flow	TSFC	T4	speed	surge-margin surrogate
631 N	1.330 kg/s	23.8 g/s	0.136 kg/(N·h)	1150 K	98.2 %	23.6 % — not validated (the surrogate fails on the NASA HECC stage)

4. Components

component	specification
Compressor	single-stage centrifugal, Ti-6Al-4V; 12 main + 12 splitter blades (splitters from 50% of the meridional length); exit backsweep -15° ; exit diameter 133.8 mm, tip speed 525 m/s (552 m/s at MCS); eye tip / hub radius 49.6 / 17.4 mm; inducer shroud relative Mach 1.29; exit width effective / physical 10.7 / 11.8 mm; blade thickness 2.24 mm at the exducer root → 0.8 mm at the tip; tip clearance 0.25 mm; boreless hub with back-face boss. η_{tt} : tool 0.817 (TurboFlow), 0.803 (NASA turbo-design, PR 4.05); fielded 0.70. Inducer choke flow $1.102 \times$ design flow (requirement 1.10). Wheel 1.082 kg (Phase 6 CAD).
Diffuser (provisional)	vaneless gap to $r_3 = 70.9$ mm (1.06 r_2), then 19-vane radial vaned diffuser to $r_4 = 90.3$ mm (R_4/R_2 1.35), vane angle $70.6^\circ \rightarrow 34^\circ$ from radial, channel width 10.7 mm; 90° bend inside r_4 into an axial deswirl annulus (30 vanes, placeholder count). Sets the engine diameter.
Combustor (provisional)	annular, casing OD 183.6 mm, annulus radii 90.3 / 22.6 mm, liner length 150 mm (0.8 $\times$ the length rule, set by rotor dynamics), reference velocity 23.6 m/s; 8 vaporiser tubes $\varnothing$ 8 mm; liner Inconel 625; η_b 0.95 (assumed), pressure loss 5 %.
Turbine	single-stage axial, IN-713LC, uncooled; 34 nozzle guide vanes / 28 rotor blades ; rotor tip radius 61.1 mm, exactly on the 350 MPa blade-root stress limit (limit 61.1 mm); rotor hub / tip radius at exit 36.7 / 61.1 mm; chord NGV / rotor 11.4 / 12.2 mm; η_{tt} tool 0.933, fielded 0.75; pressure ratio 2.60; constant-stress (Stodola) disc.
Nozzle	fixed convergent, throat area 70.8 cm ² (r_8 47.5 mm), pressure ratio 2.76 at the design point (choked), jet velocity 566 m/s; in the aircraft, a 1.09 m jetpipe leads to the tail nozzle.
Rotor & bearings (provisional)	AISI 4340 tube shaft 32×25.6 mm, journals $\varnothing$ 12 mm; two hybrid ball bearings $12 \times 28 \times 8$ mm, span 192 mm , overhung impeller (41 mm) and turbine (19 mm); damped soft supports; rotor 2.11 kg, I_p 2.13e-3 kg·m ² ; bearing DN 0.945e6 at MCS.

5. Rotor dynamics (API 684 criteria)

mode	critical speed	damping ratio ζ	amplification factor	separation margin (required)	status
1	8,335 rpm	0.189	2.64	68 % (5 %)	pass
2	14,383 rpm	0.390	1.28	45 % (0 %)	pass
3	105,571 rpm	0.017	29.09	34 % (26 %)	pass

Operating range to 78,750 rpm lies between modes 2 and 3; overall API check: **pass**. Model: ROSS beam rotor with the damped bearing supports.

6. Mass

Bottom-up model 5.70 kg $\times$ calibration 1.20 (fitted on the JetCat P400 and AMT Nike; range $\times 1.14$ -1.27) = **6.85 kg**. The calibration covers the items no model draws (flanges, fasteners, seals, harness), so the CAD is compared with the raw model.

item	model raw [kg]	Phase 6 CAD [kg]	CAD / model
impeller	0.924	1.082	1.17
diffuser + deswirl	0.389	0.199	0.51
inlet shroud + front wall	0.442	0.524	1.18
NGV vanes	0.102	0.109	1.07
NGV rings	0.138	0.140	1.01
turbine blades	0.106	0.111	1.05
turbine disc	0.280	0.281	1.00
turbine shroud	0.201	0.163	0.81
combustor liners + vaporisers	0.536	0.551	1.03
outer casing (compressor + hot section)	0.726	1.112	1.53
nozzle cones	0.101	0.104	1.02
shaft	0.577	0.424	0.74
shaft tunnel + bearing housings	0.347	0.324	0.93

item	model raw [kg]	Phase 6 CAD [kg]	CAD / model
bearings (2, hybrid)	0.050	0.063	1.26
starter motor	0.180	not drawn	
fuel manifold + igniter	0.080	not drawn	
fasteners, seals, balancing (10 %)	0.518	not drawn	
items drawn in the CAD	4.919	5.187	1.05

7. Structural status (maximum continuous speed, Ti-6Al-4V minimum yield Fty 622 MPa at ~250 °C)

location	Phase 3R analysis	Phase 6 3D FE (two meshes)	assessment
impeller exducer blade root	618 MPa incl. Kt 1.4 (plate model; sized to Fty)	578-596 MPa incl. Kt 1.4	93-96 % of Fty: on the limit
impeller inducer blade root	195 MPa (1D radial tension)	355-360 MPa (497-504 incl. Kt 1.4)	inside the limit; the 1D method misses the in-plane load path
impeller disc / hub	313 MPa mechanical, 335 MPa with thermal gradient	331-333 MPa mechanical	~76 % yield margin with thermal
impeller burst	speed ratio 1.90 (≥ 1.20 required, 14 CFR 33.27)	-	pass
blade-to-casing tip clearance	0.25 mm cold	closes 0.074 mm (mechanical)	thermal growth not included
turbine blade root	on the 350 MPa conceptual allowable (IN-713LC)	-	on the limit; life data needed

8. Installed in the aircraft

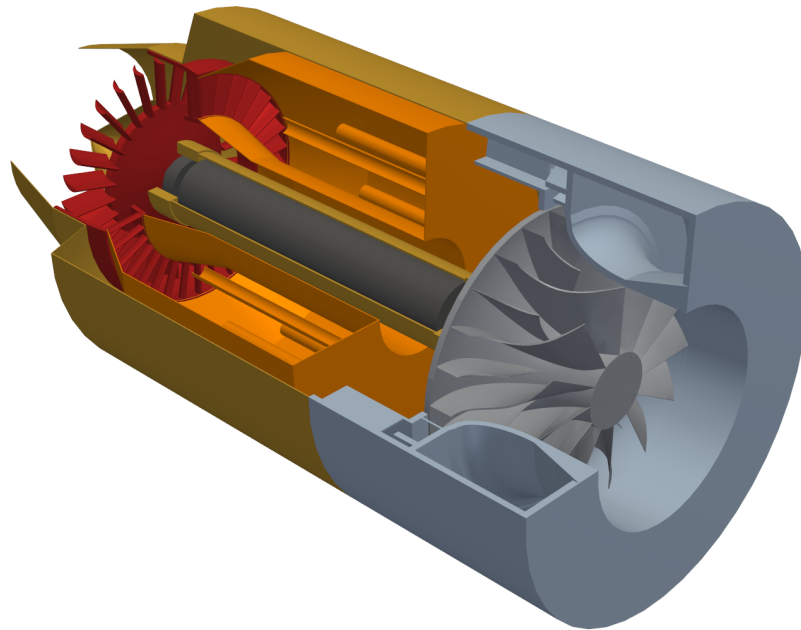
item	value
take-off gross weight / margin to 25 kg	20.09 kg / 4.91 kg
fuel (sortie + reserve + unusable)	2.98 kg; sortie 330 s
dash thrust margin (nominal / pessimistic wave drag)	+54.7 % / +28.5 %
brake release to M 1.02 at 5 km / ground roll	27.7 s / 46 m
intake / jetpipe	pitot, capture 55.1 cm ² , duct area ratio 1.42; jetpipe 1.09 m at M 0.34, friction loss 0.64 % of total pressure (not in the cycle)
engine to fuselage skin	8.7 mm radial at the area-rule waist

9. Open risks (none resolved)

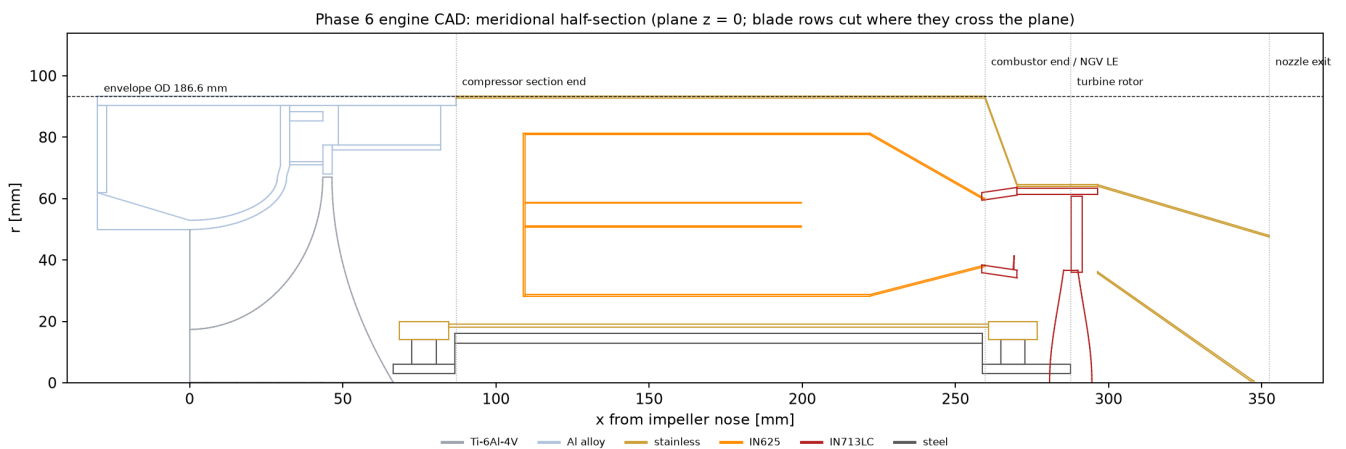
risk	status
compressor surge margin (vaned-diffuser stall)	OPEN, top risk. No validated prediction method; the peak-PR surrogate fails on the NASA HECC stage
surge margin in the idle range / start bleed	OPEN; no bleed sized or reserved
combustor shortened 20 % for rotor dynamics	OPEN; combustion performance unverified
19 diffuser vanes excite exducer mode 1 at idle	OPEN; 3D blade modes not run
exducer root and turbine blade root on their limits	confirmed in 3D (exducer); fatigue / life not assessed
inducer camber (rapid unloading, set by the 10 % choke margin)	aerodynamic loading not analysed (needs CFD)
hardware	no rig or bench test of any component

Sources: design_log.md (Phases 3R-6), docs/design_freeze.md, docs/phase6_cad.md, and the data files listed in scripts/phase6_cad/spec_sheet.py. Regenerate with .venv-cad\Scripts\python scripts\phase6_cad\spec_sheet.py.

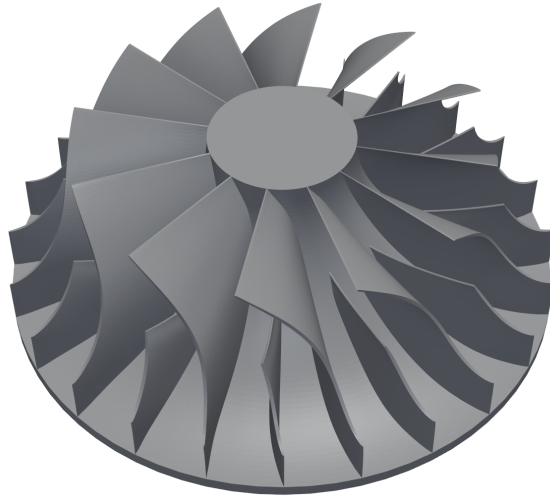
10. Geometry (Phase 6 CAD)



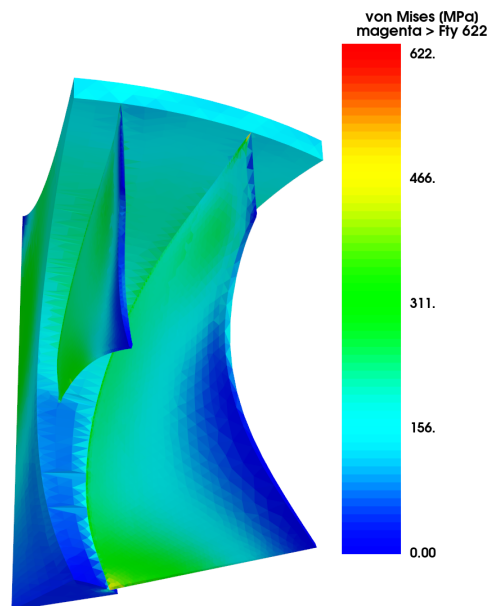
Engine cutaway: centrifugal impeller and vaned diffuser (right), annular combustor with vaporisers, single-stage axial turbine, convergent nozzle. Static parts cut open over one quadrant; the rotor is shown whole.



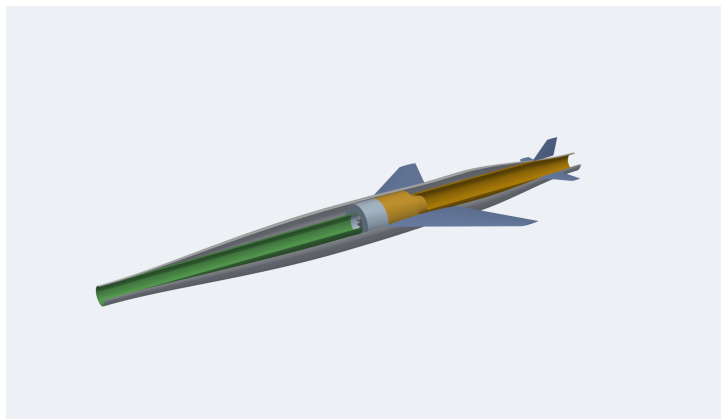
Meridional half-section of the CAD (plane $z = 0$), coloured by material.



Impeller: 12 main + 12 splitter blades, -15° backsweep, D2 134 mm, Ti-6Al-4V (1.08 kg).



3D FE, von Mises stress at maximum continuous speed on the 30° cyclic sector (magenta would mark stress above yield; the yellow spot at the blade LE root is a sharp-corner singularity of the unfilleted CAD).



Installation: pitot intake duct, engine at the area-ruled waist, jetpipe to the tail nozzle.